Nex-Gen Lumina — Lumina Bridge Setup

Describes Lumina app version 2.5.10+89 and bridge firmware v1.2.

The **Lumina Bridge** is a small device that sits on your home Wi-Fi and lets you control your lights from anywhere — the office, a vacation, the driveway. No port forwarding, no tinkering with your router. Once it's set up, remote control just works.

What you'll need

- Your **Lumina Bridge** (a small plug-in device from Nex-Gen LED LLC)
- A **USB data cable** for the bridge (it needs data transfer, not just charging)
- A computer with an internet connection
- Your **home Wi-Fi name and password**
- Your Lumina controller already installed and working on your home Wi-Fi
- Your Lumina app signed in on your phone

> **Most Nex-Gen customers have their installer set up the bridge at the end of the installation.** If yours was already configured when you got home, skip to Part 3 — you only need to verify it in the app.

How remote access works

Lumina is smart about where you are:

- **At home on your Wi-Fi:** the app talks to your lights directly — fast and local. The bridge isn't involved.
- **Away from home (cell data, a hotel, another Wi-Fi):** the app sends commands through the cloud. The bridge — always online at your house — picks them up and passes them to your lights within a couple of seconds.

Every time you open the app, Lumina runs a quick **bridge health check** — a friendly handshake that makes sure the bridge is awake and listening. The check only runs when **Remote Access is enabled** in your settings; if you only use Lumina on your home Wi-Fi, no idle pings are sent. The result shows up as a small status dot on your home screen:

Indicator	What it means
Checking	Health check is in progress
Online (green)	Bridge responded — remote control is ready
Offline (grey)	Bridge didn't respond — check power and Wi-Fi

The bridge also phones home every 30 seconds with a short status update, so you always know how long it's been running and how many commands it's processed.

> **What the bridge does and doesn't carry.** Remote commands cover the things you'd reach for while away — power, brightness, colors, patterns, and scenes. **Schedule (timer) configuration is a local-network write only.** Creating or editing a schedule while you're away saves it to your account, but it does not arm on the controller until the app is back on your home Wi-Fi. Open the Schedule tab once when you get home and it syncs itself.

Part 1 — Flash the bridge firmware

If your bridge already has firmware loaded (most do), you can skip to Part 2.

Step 1: Install PlatformIO

Install the [PlatformIO IDE extension] in VS Code. It's free, and it handles all the build and flash steps for you.

Step 2: Open the firmware project

Open the `esp32-bridge/` folder in VS Code. PlatformIO detects it automatically.

Step 3: Connect the bridge

Plug in the USB cable. If the bridge doesn't show up as a serial port:

- **Windows:** Install the CP2102 or CH340 USB driver. Check Device Manager → Ports (COM & LPT).
- **Mac/Linux:** Run `ls /dev/tty.*` or `ls /dev/ttyUSB*`.

Step 4: Build and flash

Run these PlatformIO commands from the sidebar or terminal:

```
# Build and upload the firmware
pio run -t upload

# Upload the web UI (setup page, dashboard, logo)
pio run -t uploadfs
```

> **If the flash hangs at "Connecting...":** Hold the **BOOT** button on the bridge for 3–5 seconds while the flasher is trying to connect, then let go.

Step 5: Confirm the flash worked

Open the Serial Monitor at **115200 baud**. You should see something like this:

```
╔════════════════════════════╗
║  Lumina Bridge v1.2        ║
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[FS] LittleFS mounted
[CFG] Device: Lumina-XXXX
[CFG] User ID: (not paired)
[CFG] WLED IP: 192.168.50.91:80
[WiFi] Starting AP: Lumina-XXXX
[WiFi] AP IP: 192.168.4.1
[API] HTTP server started on port 80
[Bridge] Firestore bridge module initialized
```

That's the bridge booting up and announcing itself. You're ready for Part 2.

Part 2 — Set up the bridge

The bridge runs a friendly setup wizard in a web browser. You connect to the bridge's temporary Wi-Fi, walk through three quick steps, and the bridge takes care of the rest.

Step 1: Connect to the bridge's Wi-Fi

1. On your phone or computer, look for a Wi-Fi network called **Lumina-XXXX** (the last 4 characters are unique to your bridge).
2. Connect to it — no password required.
3. A setup page should open automatically. If it doesn't, open a browser and go to `http://192.168.4.1/setup`.

Step 2: Connect the bridge to your home Wi-Fi

1. The setup page scans for available networks and lists them.
2. Tap your **home Wi-Fi** in the list.
3. Enter your Wi-Fi password and tap **Connect**.
4. Wait for the confirmation. You'll see "Connected! IP: x.x.x.x" and the wizard moves to the next step.

> **Heads-up:** The bridge only supports **2.4 GHz Wi-Fi**. If your router has separate 2.4 GHz and 5 GHz networks, pick the 2.4 GHz one. Most routers show them as two separate network names.

Step 3: Enter the bridge credentials

1. Enter the **email** and **password** for the bridge's cloud account. (Your installer will have provided these, or you'll create them in your Nex-Gen cloud console under Authentication → Users.)
2. Tap **Save & Continue**.

Step 4: Pair with your Lumina account

There are two ways to pair. **The in-app flow is easier and is what we recommend.**

Option A — Pair from the app (recommended)

1. In Lumina, go to **System** → **System Management** → **Remote Access** → **Set Up Bridge**.
2. Tap **Find Your Bridge**. Powered-on bridges on your network appear in the list within a minute or so, showing **Ready to pair**.
3. Tap your bridge, then confirm the pair. The app writes your account to the bridge for you — no user ID to copy by hand.
4. If discovery doesn't find it, open **Advanced** → **Enter Bridge IP** and type the bridge's IP directly.

> **If you see "Bridge already paired":** this bridge is paired to a different Nex-Gen account. Only continue with **Transfer to my account** if you are the authorized installer for this property — otherwise contact Nex-Gen support to transfer ownership.
>
> **Why this appears even after an account was deleted:** the bridge stores its pairing in its own onboard memory, not on the server, and it re-asserts that pairing every 30 seconds. Releasing a bridge from the server side does not free a bridge that is still powered on and online — it will simply overwrite the release on its next check-in. See *Releasing a bridge* below.

> **"This bridge firmware is outdated. Please reflash..."** means the bridge predates the in-app pairing handshake. Reflash it (Part 1), or use Option B.

Option B — Pair from the bridge's web page

1. Enter your **Lumina user ID**. Find it in the app under **System** → **System Management** → **Remote Access** — it's shown as **Your User ID**, with a **Copy User ID** button.
2. Enter your controller's local IP address (e.g., `192.168.50.91`).
3. Enter the controller port (default: `80`).
4. Tap **Pair & Finish**.

The bridge reboots and starts listening for commands from the cloud.

Part 3 — Turn on remote access in the app

Step 1: Make sure the bridge and your controller are on the same network

They need to be on the same Wi-Fi network. If you can open both in a browser from a computer on your home Wi-Fi, you're good.

Step 2: Enable remote access

1. Open the Lumina app and sign in
2. Tap **System** (gear icon) → **Remote Access**
3. While connected to your home Wi-Fi, tap **Detect Home Network** to save your Wi-Fi name
   - The first time you tap this, the app asks for **Location permission**. Required on Android (Android gates Wi-Fi network names behind location), recommended on iOS. If you decline, the app will tell you what's needed instead of failing silently.
   - Your network name is encrypted before it's saved — it never sits in plain text on the server.
4. Toggle **Enable Remote Access** on

Step 3: Verify it works

1. Close and reopen the Lumina app
2. The app automatically runs a bridge health check on startup
3. Check the bridge status dot on the home screen:
   - **Green** → bridge is online, remote access is ready
   - **Grey** → bridge didn't respond; check that it has power and is on your Wi-Fi

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Part 4 — End-to-end check

Run through this list to confirm everything is really working:

- [ ] **Bridge is powered on** and connected to home Wi-Fi (open `http://<bridge-ip>/` for its dashboard)

- [ ] **Bridge dashboard is all green** — Wi-Fi connected, cloud authenticated, user paired
- [ ] **Your controller is reachable** — open `http://<controller-ip>` in a browser
- [ ] **Lumina is signed in** to the same account the bridge is paired to
- [ ] **Home network saved** — Remote Access shows your Wi-Fi name
- [ ] **Remote access toggle is on**
- [ ] **Bridge status is green** on the home screen after an app restart
- [ ] **The real test:** turn off your home Wi-Fi on your phone (use cell data), open Lumina, toggle the lights. The command should reach the controller within a few seconds. If it does, you're fully set up.

The bridge dashboard

Any time you want to check on the bridge, open `http://<bridge-ip>/` from a browser on your home Wi-Fi. The dashboard shows:

- **Wi-Fi status** — connected or disconnected
- **Authentication** — whether the bridge is logged into the cloud
- **User paired** — whether your Lumina account is connected
- **Commands processed** — total commands the bridge has relayed
- **Errors** — total failed commands
- **Controller target** — the IP of your controller
- **Uptime** — how long the bridge has been running

From the dashboard you can also:

- **Re-run Setup** — go back to the wizard
- **Reboot Bridge** — restart the device
- **Factory Reset** — erase all settings and start fresh

Releasing a bridge — moving it to a different account

This is the step most people get wrong, so it is worth reading before you try it.

The bridge remembers its own pairing. Your account ID is written into the bridge's onboard memory during setup, and the bridge re-publishes that pairing to the cloud **every 30 seconds** while it is powered on. The cloud record is a copy; the bridge is the original.

That has one consequence that surprises people:

> **Clearing the pairing on the server does not free a live bridge.** If the bridge is plugged in and on Wi-Fi, it overwrites the release within 30 seconds and goes straight back to showing *paired* — to the old account. Deleting the old owner's Nex-Gen account does not free it either. Nothing you do from the app or the cloud will release a bridge that is still running.

The reliable way to release a bridge

Do this from a computer or phone on the same Wi-Fi as the bridge:

1. Open `http://<bridge-ip>/` for the bridge dashboard.
2. Tap **Factory Reset**.
3. The bridge erases its stored pairing, reboots, and comes back up unpaired — broadcasting its `Lumina-XXXX` setup Wi-Fi again.
4. Re-pair it to the new account using Part 2, Step 4.

If you cannot reach the dashboard, unplugging the bridge is not a substitute — it only stops the re-assertion while it is off. The moment it is plugged back in on the old network it re-claims the old account. **The reset is what actually frees it.**

> **If the bridge is already gone** — unplugged for good, discarded, or at a house you no longer service — a server-side release *does* stick, because there is nothing left to overwrite it. The problem is only ever a bridge that is still running.

Bridge LED indicators

Pattern	What it means

LED on briefly at boot	Starting up
LED blinks each poll cycle	Normal — checking for new commands
LED off between polls	Idle, waiting for the next poll

Quick reference

Detail	Value
Firmware	Lumina Bridge v1.2
Bridge Wi-Fi name	`Lumina-XXXX` (unique per device)
Setup URL (while connected to the bridge's Wi-Fi)	`http://192.168.4.1/setup`
Dashboard URL (on your home network)	`http://<bridge-ip>/`
Short-name URL	`http://lumina-xxxx.local/`
Health check timeout	15 seconds
Remote command timeout	30 seconds
Supported Wi-Fi	**2.4 GHz only** (no 5 GHz)
Remote access in the app	**System → Remote Access**

What success looks like

- The bridge dashboard at `http://<bridge-ip>/` shows green for Wi-Fi, authentication, and user paired
- Your Lumina home screen shows a green bridge status indicator after an app restart
- When you turn off home Wi-Fi on your phone and use cell data, the app still controls your lights within a couple of seconds

- The **Commands processed** counter on the bridge dashboard ticks up each time you change something from away

If something isn't working

"The bridge dashboard shows 'Not authenticated'." The bridge credentials are wrong or the account doesn't exist. Open `http://<bridge-ip>/setup` and re-enter the email and password. If you just created the account, confirm it exists in your Nex-Gen cloud console.

"The bridge dashboard shows 'Not paired'." Your Lumina user ID isn't entered. Open `http://<bridge-ip>/setup` and re-enter it. Make sure the user ID matches the one in your Lumina app under **System** → **Account**.

"Commands aren't making it to my lights."

- Check the error counter on the bridge dashboard — rising errors mean the bridge can't reach your controller.
- Confirm your controller's IP is correct and the controller itself is powered on.
- Make sure the bridge and the controller are on the same Wi-Fi network.
- If you're comfortable, open the serial monitor (115200 baud) for detailed messages.

"My commands time out from the app."

- The app waits 30 seconds for a response. If the bridge has weak Wi-Fi, move it closer to your router.
- Make sure the bridge is connected to 2.4 GHz Wi-Fi, not 5 GHz.

"The bridge won't connect to my Wi-Fi."

- The bridge only supports 2.4 GHz. Select the 2.4 GHz network name on your router.
- If the password was wrong, the bridge falls back to its setup Wi-Fi after about 15 seconds. Reconnect to the `Lumina-XXXX` network and re-enter credentials.
- Check your router's connected devices list — if the bridge is there but Lumina says it's offline, power-cycle the bridge and wait 30 seconds.

"I need to start over from scratch." Open `http://<bridge-ip>/` and tap **Factory Reset**. The bridge erases all settings and boots back into its setup Wi-Fi, ready for a fresh walkthrough.

This is also the only reliable way to move a bridge to a different account — see *Releasing a bridge*.

Still stuck? Contact Nex-Gen LED LLC support — include your bridge's Wi-Fi name (`Lumina-xxxx`) and a quick description of what the dashboard shows.

Nex-Gen Lumina — Lumina Bridge Setup — August 2026 — describes app version 2.5.10+89, bridge firmware v1.2